Lab 7: Molecular Genetics — From DNA to Protein

BIOL-1

Name: ______________________ **Date:** ______________________

Overview

Every living cell follows the same basic recipe to turn genetic information into a working molecule:

> ***DNA → RNA → Protein***

This is the **Central Dogma** of molecular biology. It means your DNA is like a master cookbook that never leaves the kitchen (the nucleus). When a protein needs to be made, the cell copies just the recipe it needs into a messenger (mRNA), which travels to the ribosome — the cell's protein-building machine.

In this lab you will **be the cell**. You'll copy DNA into mRNA (*transcription*), decode mRNA into amino acids (*translation*), and discover what happens when the recipe gets a typo (*mutations*).

Learning Objectives

By the end of this lab, you will be able to:

1. Transcribe a DNA sequence into mRNA using base-pairing rules.
2. Translate an mRNA sequence into amino acids using a codon table.
3. Predict the effects of point mutations and frameshift mutations on a protein.
4. Explain the Central Dogma in your own words.

Materials

- This worksheet
- Codon table (Part 1)
- Colored pencils (optional — great for color-coding bases!)

Part 1: The Codon Table

mRNA is read in groups of **three bases** called **codons**. Each codon specifies one amino acid (or a stop signal). The table below lists all 64 codons.

> ***How to use it:*** *Find the first base in the left column, the second base in the top row, and the third base in the right column. The intersection gives you the amino acid.*

	U	**C**	**A**	**G**	
	2nd base →				*3rd base ↓*
U	**Phe** (F)	**Ser** (S)	**Tyr** (Y)	**Cys** (C)	**U**
	Phe (F)	**Ser** (S)	**Tyr** (Y)	**Cys** (C)	**C**
	Leu (L)	**Ser** (S)	⛔ STOP	⛔ STOP	**A**
	Leu (L)	**Ser** (S)	⛔ STOP	**Trp** (W)	**G**
C	**Leu** (L)	**Pro** (P)	**His** (H)	**Arg** (R)	**U**
	Leu (L)	**Pro** (P)	**His** (H)	**Arg** (R)	**C**
	Leu (L)	**Pro** (P)	**Gln** (Q)	**Arg** (R)	**A**
	Leu (L)	**Pro** (P)	**Gln** (Q)	**Arg** (R)	**G**
A	**Ile** (I)	**Thr** (T)	**Asn** (N)	**Ser** (S)	**U**
	Ile (I)	**Thr** (T)	**Asn** (N)	**Ser** (S)	**C**
	Ile (I)	**Thr** (T)	**Lys** (K)	**Arg** (R)	**A**
	🟢 **Met** (M) START	**Thr** (T)	**Lys** (K)	**Arg** (R)	**G**
G	**Val** (V)	**Ala** (A)	**Asp** (D)	**Gly** (G)	**U**
	Val (V)	**Ala** (A)	**Asp** (D)	**Gly** (G)	**C**
	Val (V)	**Ala** (A)	**Glu** (E)	**Gly** (G)	**A**
	Val (V)	**Ala** (A)	**Glu** (E)	**Gly** (G)	**G**

> ***Key:***
>
> - 🟢 ***AUG*** = *Start codon (also codes for Methionine)*
> - ⛔ ***UAA, UAG, UGA*** = *Stop codons (no amino acid added — translation ends)*

Quick Practice

Try decoding these three codons before moving on:

Codon	1st base	2nd base	3rd base	Amino Acid
GCA	G	C	A	_____
UUU	U	U	U	_____
AUG	A	U	G	_____

Part 2: Transcription & Translation

Background

Think of transcription like photocopying one page from a master binder — the original DNA stays safe in the nucleus while the mRNA copy travels to the ribosome.

Base-pairing rules for transcription (DNA → mRNA):

DNA base	mRNA base
A	U
T	A
G	C
C	G

Your Gene

DNA template strand:

3′ — T A C · C G T · A C G · T C G · G G T · G A C · A T T — 5′

1. Transcribe — Write the mRNA sequence. Group into codons (groups of 3).

2. Translate — Use the codon table to convert each codon into an amino acid. Write the amino acid chain.

3. What was your first codon? What amino acid does it code for, and why is that codon special?

4. How did you know to stop translating? What was the stop codon?

Part 3: Mutation Detective

Mutations are changes in the DNA sequence — like typos in a recipe. Some are harmless; others can ruin the dish. Let's investigate.

Go back to the **original DNA template strand** from Part 2.

Scenario A: Substitution (Point Mutation)

The **10th base** (the first `C` in the third codon `ACG`) changes to a **C → G**.

Mutated DNA template: `3′ — T A C · C G T · A G G · T C G · G G T · G A C · A T T — 5′`

5. Transcribe and translate the mutated sequence.

- Mutated mRNA:
- Amino acid chain:

6. Did the amino acid chain change? Classify the mutation as *silent* (no change), *missense* (different amino acid), or *nonsense* (creates a premature stop).

Scenario B: Insertion (Frameshift Mutation)

Return to the **original DNA template**. An extra **G** is inserted right after the 3rd base (after the `C` in `TAC`).

Mutated DNA template: `3′ — T A C · G C G · T A C · G T C · G G G · T G A · C A T · T — 5′`

7. Transcribe and translate the frameshift-mutated sequence.

- Mutated mRNA:
- Amino acid chain:

8. Compare the frameshift protein to your original. How many amino acids changed? Why is this type called a "frameshift"?

9. Which mutation was more damaging — the substitution (A) or the frameshift (B)? Explain your reasoning.

Part 4: Big Picture

10. Complete the Central Dogma:

DNA → ___ → ___

11. Where in a eukaryotic cell does transcription occur?

12. Where does translation occur?

13. What enzyme copies DNA into mRNA (carries out transcription)?

14. What cellular structure reads mRNA and builds proteins (carries out translation)?

15. In your own words: How can a single change in someone's DNA affect their physical traits? Trace the path from gene → protein → trait.